

AI Capstone: Milestone 1 Check-in

Due: Week 3

Focus: Solidifying the project scope, understanding the data landscape, planning the technical approach, and establishing a clear project roadmap.

Requirements & Discussion Points:

Students should be prepared to discuss and informally present the following:

1. Refined Project Charter:

- Any modifications made to the initial project idea (scope adjustments, clarified objectives).
- Clear articulation of the problem they are solving and the project's goal.
- Confirmation of In-Scope and Out-of-Scope items.

2. Data Acquisition & Initial Exploration:

- What data sources have been identified and accessed? (Show links, describe datasets).
- What are the initial findings from exploring the data? (e.g., data formats, key fields, potential quality issues, volume).
- What is the plan for data cleaning and preprocessing?
- If using simulated data, what is the plan for generation?

3. Proposed Technical Approach:

- What specific ML models, algorithms, or techniques are being considered? Why?
- What technologies (programming languages, libraries, frameworks) will be used? (e.g., Python, TensorFlow/PyTorch, Scikit-learn, Flask/FastAPI, Docker, specific cloud services). Justify choices.
- Initial thoughts on system architecture (e.g., data flow, API structure, dashboard components).

4. Project Plan & Team Roles:

- Detailed task breakdown for the *next 3 weeks* (leading to Milestone 2).
- Initial assignment of roles and responsibilities within the group.
- Identification of potential risks or roadblocks and initial mitigation ideas.

5. Questions & Challenges:

- What questions do you have? What challenges are you anticipating?

Marking Scheme (Informal Discussion-Based Assessment - Total 15 points)

- **1. Clarity of Project Definition (3 points):**
 - (0-1) Scope/goals are vague or unrealistic.
 - (2) Scope/goals are mostly clear but need minor refinement.
 - (3) Scope/goals are well-defined, clear, and seem achievable.
- **2. Data Understanding & Plan (3 points):**
 - (0-1) Data sources unclear or initial exploration is minimal; weak plan.
 - (2) Data sources identified, some exploration done; basic cleaning/prep plan exists.
 - (3) Clear understanding of data sources, good initial exploration insights; solid plan for cleaning/prep/simulation.
- **3. Technical Approach & Justification (3 points):**
 - (0-1) Technology choices seem arbitrary or inappropriate; approach unclear.
 - (2) Possible technology choices and approach, but justification could be stronger.
 - (3) Well-reasoned technology choices and technical approach clearly articulated.
- **4. Project Planning & Risk Awareness (3 points):**
 - (0-1) Next steps unclear; minimal task breakdown or risk identification.
 - (2) Basic plan for next milestone exists; some thought given to risks.
 - (3) Detailed, realistic plan for next 3 weeks; roles considered; key risks identified with mitigation ideas.
- **5. Professionalism & Engagement (3 points):**
 - (0-1) Group seems unprepared, disengaged, or struggles to articulate progress.
 - (2) Group is generally prepared and engaged, answers questions reasonably well.
 - (3) Group is well-prepared, actively engaged, communicates clearly, asks

thoughtful questions.

Feedback: Verbal feedback will be provided immediately during/after the discussion, focusing on strengths and areas for improvement leading into the next milestone.